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8 MAY 1964
SUPERSEDING
MIL-C-9877A
1 SEPTEMBER 1962

MILITARY SPECIFICATION

CARDS, APERTURE

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers aperture cards (see 6.2.1) for mounting microfilm images of engineering documents prepared by the Department of Defense or contractors.

1.2 **Classification.** Aperture cards shall be of the following types and classes:

Type I—Cold Seal, Pressure Sensitive Microfilm Carrier.
Class 1—Microfilm Carrier on Face.
Class 2—Microfilm Carrier on Back.
Type II—Suspension Microfilm Carrier.
Class 1—Microfilm Insertion Opening on Back.
Class 2—Microfilm Insertion Opening on Face.
Type III—Heat Sensitive Microfilm Carrier.
Class 1—Microfilm Carrier on Face.
Class 2—Microfilm Carrier on Back.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

G-C-116 —Cards; Tabulating.
PPP-B-636—Boxes, Fiber.
UU-P-31 —Paper, General Specifications and Methods of Testing.

MILITARY

MIL-P-116 —Preservation, Methods of.
MIL-M-9868 —Microfilming of Engineering Documents, 35mm, Requirements for.
MIL-L-10547 —Liners, Case, Waterproof.

STANDARDS

MILITARY

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.
MIL-STD-804—Formats and Coding of Tabulating and Aperture Cards for Engineering Data Micro-reproduction System.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

FSC EDMS

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3.1 Preproduction sample. When required by the procuring activity, specified quantities of aperture cards of each type, class, and style on contract shall be furnished by Government inspection and approval.

3.2 General requirements.

3.2.1 *Card stock.* Card stock shall be Type I per Specification G–C–116. Color of card stock and card size shall be in accordance with Standard MIL–STD–804.

3.2.1.1 *Edge coating.* The top and bottom edges of the card shall be thoroughly coated. The coating shall be composed of 25 percent vinyl acetate and 75 percent toluol.

3.2.2 *Card formats.* The formats of the aperture cards shall be in accordance with Standard MIL–STD–804. Card corner cut and card color striping shall be in accordance with Standard MIL–STD–804.

3.3 Detail requirements.

3.3.1 *Type I aperture cards.*

3.3.1.1 *Aperture and microfilm carrier dimensions.* The location and size of the aperture and of the adhesive-coated microfilm carrier shall be in accordance with figure 1. The adhesive-coated microfilm carrier shall be protected by a protective sheet.

3.3.1.2 *Classes.* As specified by the procuring activity, Type I aperture cards shall be furnished as either Class 1 or Class 2. Class 1 cards shall have the adhesive-coated microfilm carrier on the face of the aperture card. Class 2 cards shall have the adhesive-coated microfilm carrier on the back of the aperture card.

3.3.1.3 *Removal.* The adhesion of the adhesive-coated microfilm carrier to the card stock shall pass the 20 gram weight test of 4.6.3.

3.3.1.4 *Release.* The release of the protective sheet from the adhesive-coated microfilm carrier shall pass the release test of 4.6.4.

3.3.1.5 *Adhesion.* The adhesion of the frame of microfilm to the adhesive-coated microfilm carrier shall pass the 10 gram weight test of 4.6.5.

3.3.1.6 *Nonblocking.* The nonblocking characteristics of Type I aperture cards shall pass the fanning test of 4.6.6.

3.3.1.7 *Thickness.* The total thickness of the laminate of card plus microfilm carrier shall be not more than .0002 inch greater than the thickness of the card alone.

3.3.1.8 *Proprietary item.* Type I aperture cards are a proprietary item. The Government has a royalty-free license under these patents for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

Patent number	*Expiration date*
2,511,859	20 June 1967.
2,512,106	20 June 1967.
2,587,022	26 February 1969.

3.3.2 *Type II aperture cards.*

3.3.2.1 *Aperture and microfilm carrier dimensions.* The location and size of the aperture and of the suspension microfilm carrier shall be in accordance with figure 2.

3.3.2.2 *Classes.* As specified by the procuring activity, Type II aperture cards shall be furnished as either Class 1 or Class 2. Class 1 cards shall have the microfilm insertion opening on the back of the aperture card. Class 2 cards shall have the microfilm insertion opening on the face of the aperture card.

3.3.2.3 *Suspension microfilm carrier.* The suspension microfilm carrier shall be of static-free, optically clear material that will not adhere to itself after being bonded to the card.

3.3.2.4 *Bond.* The bonding agent shall cause a complete linkage between the fibers of the card stock, and by a chemical physical action etch the suspension microfilm carrier, only where contact with the bonding agent is present, so as to form a continuous bond between the two materials. The bond shall pass the 20 gram weight test of 4.7.2.

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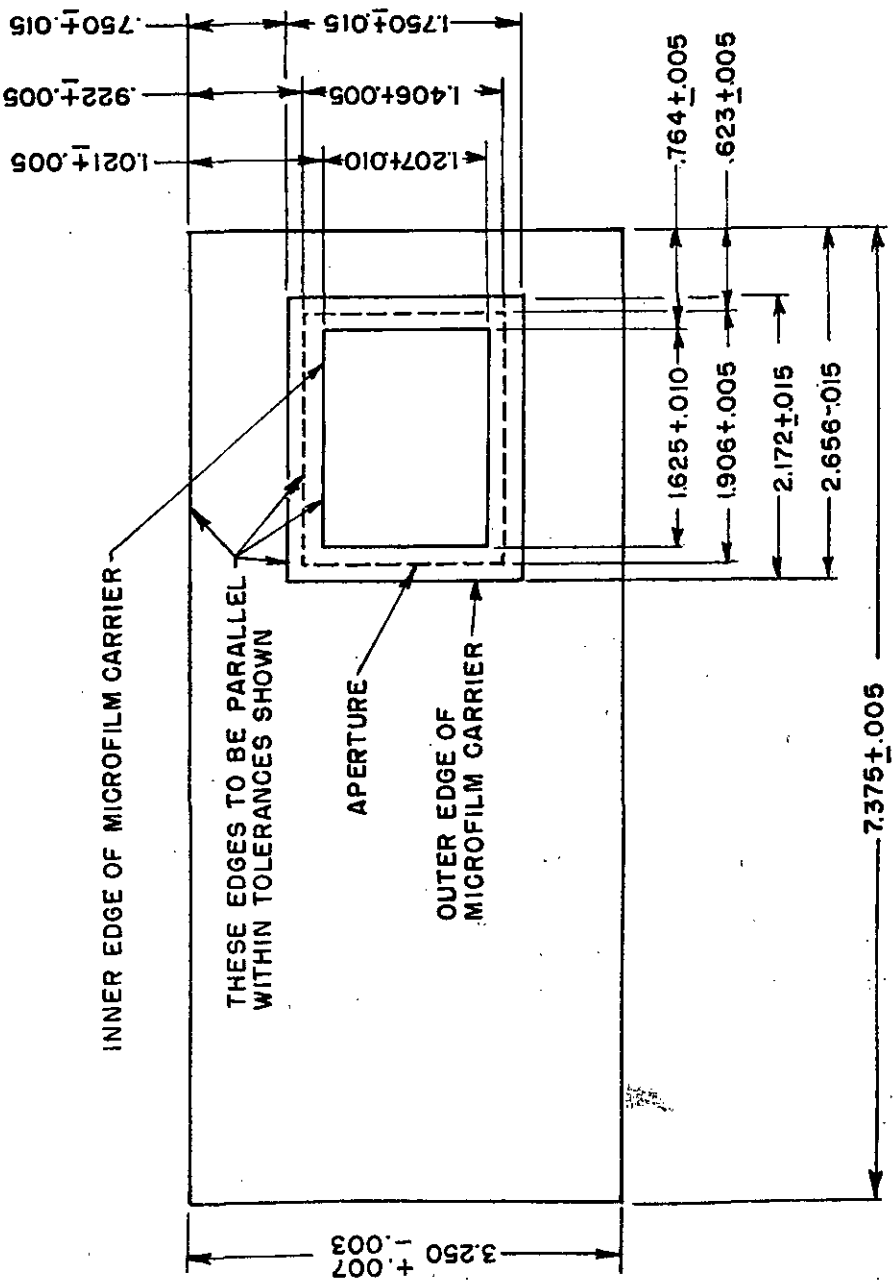


FIGURE 1. *Aperture and microfilm carrier dimensions for type I and type III card.*

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3.3.2.5 *Nonblocking.* The nonblocking characteristics of Type II aperture cards shall pass the fanning test of 4.6.6.

3.3.2.6 *Retention.* The frame of microfilm shall remain in place between the suspension microfilm carriers, with no part of the frame projecting through the microfilm insertion opening, when subjected to the retention test of 4.7.3.

3.3.2.7 *Thickness.* The total thickness of the laminate of card plus two thicknesses of suspension microfilm carrier shall be not more than .0010 inch greater than the thickness of the card alone.

3.3.2.8 *Proprietary item.* Type II aperture cards are a proprietary item. The Government does not have a royalty-free license under these patents:

Patent number	*Expiration date*
U.S. Serial # 16,727	Patent Pending.

3.3.3 *Type III aperture cards.*

3.3.3.1 *Aperture and microfilm carrier dimensions.* The location and size of the aperture and of the heat sensitive microfilm carrier shall be in accordance with figure 1.

3.3.3.2 *Classes.* As specified by the procuring activity, Type III aperture cards shall be furnished as either Class 1 or Class 2. Class 1 cards shall have the heat sensitive microfilm carrier on the face of the aperture card. Class 2 cards shall have the heat sensitive microfilm carrier on the back of the aperture card.

3.3.3.3 *Removal.* The adhesion of the heat sensitive microfilm carrier to the card stock shall pass the 30 gram weight test of 4.6.3.

3.3.3.4 *Adhesion.* The adhesion of the frame of microfilm to the heat sensitive microfilm carrier shall pass the 15 gram weight test of 4.6.5.

3.3.3.5 *Nonblocking.* The nonblocking characteristics of Type III aperture cards shall pass the fanning test of 4.6.6.

3.3.3.6 *Thickness.* The total thickness of the laminate of card plus microfilm carrier shall be not more than .0007 inch greater than the thickness of the card alone.

3.3.3.7 *Proprietary item.* Type III aperture cards are a proprietary item. The Government does not have a royalty-free license under these patents:

Patent number	*Expiration date*
U.S. Serial # 81,325	Patent Pending.

3.4 Workmanship. Aperture cards shall be clean, curl-free, free from nicks, folds, bent corners, torn corners, or other card damage that might affect punched card accounting machine processing.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Acceptance of items before preparation for delivery (see 4.4).
(c) Acceptance inspection of preparation for delivery (see 4.5 and sec. 5).

4.3 Preproduction inspection. Preproduction inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of all the inspections in table I for Type I and III aperture cards and all of the inspections in table II for Type II aperture cards, as well as such other tests and inspections as are required to determine compliance with specified requirements.

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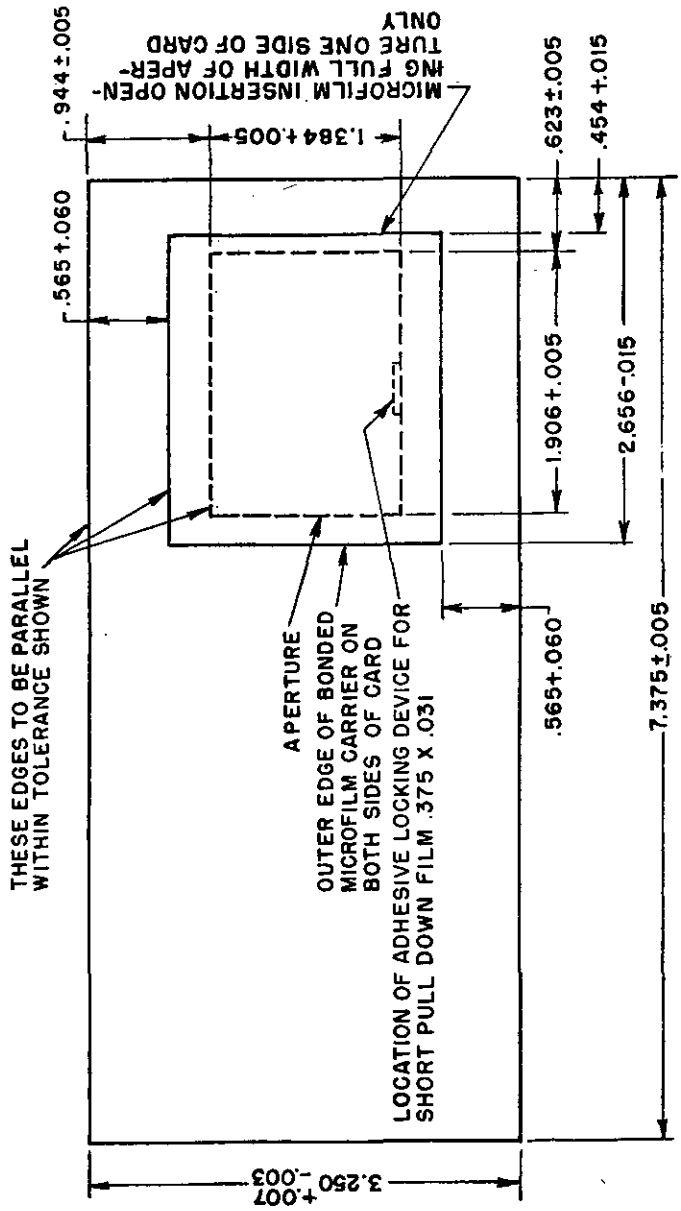


FIGURE 2. *Aperture and microfilm carrier dimensions for type II card.*

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4.4 Acceptance inspection of item before preparation for delivery. The contractor, to demonstrate compliance with specified requirements, shall perform the inspection specified below. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government —at its discretion—may perform all or part of the inspection specified herein, to verify the contractor's compliance with specified requirements. (The amount of such verification inspection will depend on the extent to which the contractor's inspection procedures and records insure that the product on contract meets the specified procurement inspection (also see 6.3).)

4.4.1 *Inspection of Type I cards.* This inspection (including sampling) shall conform to table I and the ordinary inspection procedures of Standard MIL–STD–105.

TABLE I. *Inspection of type I cards*

Inspection	Requirement paragraph	Inspection paragraph	AQL
Dimensions	3.2.1	4.6.1	Major .40 dphu
	3.3.1.1	4.6.1	Minor 1.0 dphu
Removal	3.3.1.3	4.6.3	
Release	3.3.1.4	4.6.4	
Adhesion	3.3.1.5	4.6.5	.40 dphu
Nonblocking	3.3.1.6	4.6.6	
Thickness	3.3.1.7	4.6.7	3 aperture cards per lot
Visual	3.4	4.6.8	.65 dphu

4.4.2 *Inspection of Type II cards.* This inspection (including sampling) shall conform to table II and the ordinary inspection procedures of Standard MIL–STD–105.

TABLE II. *Inspection of type II cards*

Inspection	Requirement Inspection	Inspection paragraph	AQL
Dimensions	3.2.1	4.7.1	Major .40 dphu
	3.3.2.1	4.7.1	Minor 1.0 dphu
Bond	3.3.2.4	4.7.2	
Nonblocking	3.3.2.5	4.6.6	.40 dphu
Retention	3.3.2.6	4.7.3	
Thickness	3.3.2.7	4.7.4	3 aperture cards per lot
Visual	3.4	4.7.5	.65 dphu

4.4.3 *Inspection of Type III cards.* This inspection (including sampling) shall conform to table III and the ordinary inspection procedures of Standard MIL–STD–105.

TABLE III. *Inspection of type III cards*

Inspection	Requirement paragraph	Inspection paragraph	AQL
Dimensions	3.2.1	4.6.1	Major .40 dphu
	3.3.3.1	4.6.1	Minor 1.0
Removal	3.3.3.3	4.6.3	
Adhesion	3.3.3.4	4.6.5	.40 dphu
Nonblocking	3.3.3.5	4.6.6	
Thickness	3.3.3.6	4.6.7	3 aperture cards per lot
Visual	3.4	4.6.8	.65 dphu

4.5 Acceptance inspection for preparation for delivery. The contractor, to demonstrate compliance with specified requirements, shall perform the inspection specified by 4.5.1 and 4.5.2. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government—at its discretion—may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements (see 6.3).

4.5.1 *Preservation and packaging.* Preservation and packaging shall be inspected in accordance with Specification MIL–P–116. Classification of defects shall be as shown on table IV.

4.5.2 *Packing and marking of exterior containers.* Packing and marking of exterior containers shall be given visual inspection in accordance with table V, to determine conformance to section 5 and the approved process sheet (see contract). This inspection shall conform to the special procedure for small-sample inspection of Standard MIL–STD–105. Inspection level L–7 shall be used for normal inspection and L–5 for reduced inspection. The reduced inspection procedure shall be R–1.

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Note. Other statistical sampling procedures may be employed if approved by the contracting officer.

TABLE IV. *Classification of defects for preservation, packaging and marking*

Major	Minor
1. Use of improper or defective material.	1. Item not properly blocked or braced within the unit package to prevent movement.
2. Quantity in unit package not as specified.	2. Packaging material damaged.
3. Incorrect packaging method applied.	3. Any item of required marking information other than 4 through 6 listed under major defects omitted, incorrect or illegible.
4. Contract or order number omitted, incorrect, or illegible.	
5. Identification omitted, incorrect, or illegible.	
6. Marking of quantity of items in package incorrect or illegible.	

(a) *Major defects:* For sample sizes up to 10 inclusive, as determined by reference to Standard MIL–STD–105, each sample size as given shall be used and for these sample sizes the acceptance number shall be zero. For sample sizes greater than 10 the acceptance number shall be determined by the AQL of 4.0 percent defective.

(b) *Minor defects:* For sample sizes up to 5 inclusive, as determined by reference to Standard MIL–STD–105, each sample size as given shall be used and for these sample sizes the acceptance number shall be zero. For sample sizes greater than 5 the acceptance number shall be determined by the AQL of 10.0 percent defective.

4.5.2.1 *Inspection lot.* A lot for visual inspection of the pack shall be all completed packs which are identical and will be submitted for acceptance at one time.

4.5.2.2 *Procedure in case of failure.* When an inspection lot fails, the contractor shall immediately investigate the cause of failure and take corrective action to assure that subsequent lots do not contain the same defect or defects.

TABLE V. *Classification of defects for packing and marking*

Major	Minor
1. Use of improper or defective material.	1. Defective taping or sealing of carton.
2. Quantity in pack not as specified.	2. Any other box defect which may be considered minor by definition of Standard MIL–STD–105.
3. Gross weight in excess of specified amount.	3. Any item of required marking information other than 8 through 15 listing under major defects omitted, incorrect, or illegible.
4. Box closure not as specified.	
5. Type, grade class, and style of the shipping container not as specified.	
6. Items not adequately blocked, braced, or cushioned within the shipping container to prevent movement or damage.	
7. Shipping documents or packing list omitted.	
8. Contract or order number omitted, incorrect, or illegible.	
9. Identification omitted, incorrect, or illegible.	
10. Marking of quantity of items in pack omitted, incorrect, or illegible.	
11. Destination marking omitted, incorrect, or illegible.	
12. Service designation (color marking) omitted.	
13. Special marking or labeling (when required) omitted.	
14. Oversea code marking omitted, incorrect, or illegible.	
15. Unsealed carton.	

4.5.2.3 *Disposition of nonconforming product.* Disposition of nonconforming product shall conform to Standard MIL–STD–105. Such product shall be suitably tagged or identified to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted.

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4.6 Inspection procedures for Type I and Type III cards.

4.6.1 *Dimensional inspection of Type I and Type III cards.* The card dimensions listed below shall be gaged or measured to determine conformance to requirements of 3.2.1 and 3.3.1.1. When a listed dimension is not within specified limits, it shall be considered a major or minor defect as indicated.

(a) 3.250 inches plus .007 inch, minus .003 inch (major).
(b) 7.375 inches plus or minus .005 inch (major).
(c) Thickness .0070 inch plus or minus .0004 inch (major).
(d) 1.021 inches plus or minus .005 inch (major).
(e) 1.207 inches plus .010 inch, minus .000 inch (major).
(f) .764 inch plus or minus .005 inch (major).
(g) 1.625 inches plus .010 inch, minus .000 inch (major).
(h) .922 inch plus or minus .005 inch (major).
(i) 1.406 inches plus .005 inch, minus .000 inch (major).
(j) .623 inch plus or minus .005 inch (major).
(k) 1.906 inches plus .005 inch, minus .000 inch (major).
(l) Parallelism of 3 edges within plus or minus .005 inch (major).
(m) .750 inch plus or minus .015 inch (minor).
(n) 1.750 inches plus or minus .015 inch (minor).
(o) 2.656 inches plus .000 inch, minus .015 inch (minor).
(p) 2.172 inches plus or minus .015 inch (minor).
(q) Upper left corner cut ¼ inch along top edge of card (minor).
(r) Upper left corner cut ⅜ inch along left edge of card (minor).

4.6.2 *Conditioning.* Conditioning of the sample shall not be started until at least 48 hours have elapsed since the cards were manufactured. The sample shall be conditioned, loosely assembled, at a temperature of 70 degrees F. plus or minus 3 degrees F., and at a relative humidity of 50 percent, plus or minus 10 percent, for the specified period.

4.6.3 *Removal test for Type I and Type III cards.* The sample shall be conditioned in accordance with 4.6.2 for a period of 6 hours before testing. After conditioning, the peel of the microfilm carrier from the card stock shall be not more than one-half inch in one minute with a 20 gram weight for Type I cards, or a 30 gram weight for Type III cards, suspended on the microfilm carrier when tested in accordance with the following.

4.6.3.1 The equipment for this test shall include (a) a card suspension rack with adjustable scale, (b) the appropriate calibrated weight with alligator clip, and (c) a stop watch. The card suspension rack shall be a vertically arranged clip board in which the center portion has been cut out to allow unimpeded drop of the weight during the test. The scale shall be one inch wide metal strip mounted on a vertical post to allow for vertical adjustment. The weight shall be brass, attached by a solid wire, to a small alligator clip. The body of the weight shall be cylindrical and shall have provision for calibration through the addition or removal of solder. Calibration shall be to the nearest .001 gram of the weight specified for the test. The lower end of the weight shall be flat and at a right angle with the side of the cylinder. Cards, after visual inspection and the specified conditioning, shall be tested as follows.

4.6.3.1.1 Slit the full length of the microfilm carrier, parallel to the long edge of the aperture, adjacent to the edge of the aperture.

4.6.3.1.2 Peel back the microfilm carrier, from the corner of the aperture towards the 80 column end of the card, approximately ¼ to ⅜ inch.

4.6.3.1.3 Suspend the prepared card in the rack with the aperture down, the side under

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test away from the board, and the top edge of the card against the base of the card clip.

4.6.3.1.4 Attach the weight, by means of the alligator clip, to the peeled back portion of the microfilm carrier.

4.6.3.1.5 Gently release the weight and adjust the scale so that the lower end of the weight will be in line with the top edge of the scale.

4.6.3.1.6 When the weight is at rest and in line with the top edge of the scale, start the stop watch.

4.6.3.1.7 Observe the position of the weight after one minute has elapsed. The item under test shall be considered a defect if the weight has dropped so that its lower end has passed the bottom edge of the scale.

4.6.3.2 Cards subjected to this test shall not be delivered on contract or order.

4.6.4 *Release test for Type I cards.* The sample shall be conditioned in accordance with 4.6.2 for a period of 6 hours before testing. After conditioning, the strength of the bond of the protection sheet to the adhesive-coated microfilm carrier shall be sufficient to maintain the assembly intact when the card is caused to roll around a mandril of 1⅜-inch diameter. The long side of the card shall be parallel to the axis of the mandril. The face of Class 1 cards and the back of Class 2 cards shall be against the mandril. Cards subjected to this test shall not be delivered on contract or order.

4.6.5 *Adhesion test for Type I and Type III cards.* Frames of processed Type I microfilm in accordance with Specification MIL–M–9868 shall be mounted in each aperture card in the sample in accordance with the aperture card manufacturer's instructions. The sample then shall be stored at a temperature of 70 degrees F., plus or minus 3 degrees F., and at a relative humidity of 50 percent, plus or minus 10 percent, for at least 6 hours with a weight of eleven pounds placed on the aperture area. The sample shall then be conditioned in accordance with 4.6.2 for a period of 6 hours. After conditioning, the peel of the microfilm carrier from the frame of microfilm shall be not more than one-half inch in one minute with a 10 gram weight for Type I cards, or a 15 gram weight for Type III cards suspended on the adhesive-coated microfilm carrier when tested in accordance with 4.6.3.1 and subparagraphs thereto. Cards subjected to this test shall not be delivered on contract or order.

4.6.6 *Nonblocking test for aperture cards.* The sample for this test shall consist of a consecutive group of cards of the sample size "n" in accordance with Standard MIL–STD–105 selected without regard to quality. The sample shall be conditioned in accordance with 4.6.2 for a period of 24 hours before testing. The sample then shall be stored for a period of 24 hours at a temperature of 115 degrees F., plus or minus 5 degrees F., under a weight of 11 pounds placed on the aperture area. It shall then be reconditioned in accordance with 4.6.2 for a period of 6 hours. After reconditioning, the sample then shall be fanned and compared for ease of fanning with the fanning of a like number of tabulating cards. Each individual card shall separate from its adjacent card without card-to-card adhesion due to aperture card manufacturer. Every two cards that adhere together shall be counted as a defect. Cards subjected to this test shall not be delivered on contract or order.

4.6.7 *Thickness test for Type I and Type III cards.* The thickness shall be measured by use of the apparatus described in Specification UU–P–31, Method 173, Section 3. Each card in the sample shall be measured at a place selected without regard to quality. Each sample card shall be placed between the jaws of the instrument and the pressure foot lowered gently upon the surface to be measured. The laminate shall be measured with the centerline of the anvil in line with the center of the laminate of card plus microfilm carrier. The thickness of the card shall be measured in the area adjacent to the measurement of the laminate, with the anvil making full contact with the card stock. If the thickness of

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the laminate exceeds the thickness of the card stock by more than .0002 inch for Type I cards or .0007 for Type III cards, it shall be counted as a defect. If there are two or more defects for this test in an inspection lot, the lot shall be rejected.

4.6.8 *Visual inspection for Type I and Type III cards.* The cards shall be inspected for the defects listed below:

(a) Card edge is visibly nicked.
(b) Card is torn.
(c) Card is folded or creased.
(d) Corner of card is bent.
(e) Microfilm carrier is on wrong side of card.
(f) Microfilm carrier is pleated, wrinkled, or torn.
(g) Area of microfilm carrier is not coated with adhesive.
(h) Inner edge microfilm carrier is not flat and tight against protective sheet. (When applicable).
(i) Segment of microfilm carrier is missing.
(j) Foreign material is stuck to adhesive area of microfilm carrier. (When applicable).
(k) Corner of aperture in card is not a sharp right angle.
(l) Protective sheet is incorrectly located or is lapped on surface of card.
(m) Format of card is incorrect.
(n) Corner cut is incorrect.
(o) Color of card is incorrect.
(p) Color of stripe is incorrect.
(q) Stripe is in wrong position.

4.7 Inspection procedures for Type II cards.

4.7.1 *Dimensional inspection of Type II cards.* The card dimensions listed below shall be gaged or measured to determine conformance to the requirements of 3.2.1 and 3.3.2.1. When a listed dimension is not within specified limits, it shall be considered a major or minor defect as indicated.

(a) 3.250 inches plus .007 inch, minus .003 inch (major).
(b) 7.375 inches plus or minus .005 inch (major).
(c) Thickness .0070 inch plus or minus .0004 inch (major).
(d) .944 inch plus or minus .005 inch (major).
(e) 1.384 inches plus .005 inch, minus .000 inch (major).
(f) .623 inch plus or minus .005 inch (major).
(g) 1.906 inches plus .005 inch, minus .000 inch (major).
(h) .565 inch plus .060 inch, minus .000 inch minor (2 places).
(i) .454 inch plus .015 inch, minus .000 inch (minor).
(j) 2.656 inches plus .000 inch, minus .015 inch (minor).
(k) Width of microfilm carrier opening 1.384 inches plus .005 inch, minus .000 inch (major).
(l) Upper left corner cut ¼ inch along top edge of card (minor).
(m) Upper left corner cut ⅜ inch along left edge of card (minor).

4.7.2 *Bond test for Type II cards.* The peel of the suspension microfilm carrier from the card stock shall be not more than one-half inch in one minute with a 20 gram weight suspended on the suspension carrier when tested in accordance with the following:

4.7.2.1 The equipment for this test shall include (a) a card suspension rack with adjustable scale, (b) the appropriate calibrated weight with alligator clip, and (c) a stop watch. The card suspension rack shall be a vertically arranged clip board in which the center portion has been cut out to allow unimpeded drop of the weight during the test. The scale shall be a one inch wide metal strip mounted on a vertical post to allow for vertical adjustment. The weight shall be brass, attached by a solid wire, to a small alligator clip. The body of the weight shall be cylindrical and shall have provision for calibration through the addition or removal of solder. Calibration shall be to the nearest .001 gram

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of the weight specified for the test. The lower end of the weight shall be flat and at a right angle with the side of the cylinder. Cards, after visual inspection and the specified conditioning, shall be tested as follows:

4.7.2.1.1 Slit the full length of the microfilm, parallel to the long edge of the aperture, adjacent to the edge of the aperture.

4.7.2.1.2 Slit the microfilm carrier to the specified width.

4.7.2.1.3 Peel back the microfilm carrier, from the corner of the aperture towards the 80 column end of the card, approximately ¼ to ⅜ inch.

4.7.2.1.4 Suspend the prepared card in the rack with the aperture down, the side under test away from the board, and the top edge of the card against the base of the card clip.

4.7.2.1.5 Attach the weight, by means of the alligator clip, to the peeled back portion of the microfilm carrier.

4.7.2.1.6 Gently release the weight and adjust the scale so that the lower end of the weight will be in line with the top edge of the scale.

4.7.2.1.7 When the weight is at rest and in line with the top edge of the scale, start the stop watch.

4.7.2.1.8 Observe the position of the weight after one minute has elapsed. The item under test shall be considered a defect if the weight has dropped so that its lower end has passed the bottom edge of the scale.

4.7.2.2 Cards subjected to this test shall not be delivered on contract or order.

4.7.3 *Retention test for Type II cards.* A frame of microfilm shall be mounted in each card of the sample. The sample shall then be conditioned in accordance with 4.6.2 for a period of 6 hours before testing. After conditioning, the frame of microfilm shall remain in place between the suspension microfilm carriers, with no part of the frame projecting through the microfilm insertion opening when the card is caused to roll around a mandril of 1⅜ inch diameter. The long side of the card shall be parallel to the axis of the mandril. The face of Class 1 cards and the back of Class 2 cards shall be against the mandril. Cards subjected to this test shall not be delivered on contract or order.

4.7.4 *Thickness test for Type II cards.* The thickness of the laminate of card plus two thicknesses of suspension microfilm carrier shall be measured on each card in the sample at a place selected without regard to quality along a side of the aperture opening in the card. If the thickness of the laminate exceeds the thickness of the card stock by more than .0010 inch, it shall be counted as a defect. If there are two or more defects for this test in an inspection lot, the lot shall be rejected.

4.7.5 *Visual inspection of Type II cards.* The cards shall be inspected for the defects listed below:

(a) Card edge is visibly nicked.
(b) Card is torn.
(c) Card is folded or creased.
(d) Corner of card is bent.
(e) Microfilm insertion opening is on wrong side of card.
(f) Suspension microfilm carrier is pleated, wrinkled, or torn.
(g) Suspension microfilm carrier is not static free.
(h) Suspension microfilm carrier is not optically clear.
(i) Suspension microfilm carriers adhere together in card.
(j) Bond of suspension microfilm carrier to stock is not continuous.
(k) Foreign material is present between suspension microfilm carrier.
(l) Corner of aperture card is not a sharp right angle.
(m) Format of card is incorrect.
(n) Corner cut is incorrect.
(o) Color of card is incorrect.
(p) Color of stripe is incorrect.
(q) Stripe is in wrong position.

5. PREPARATION FOR DELIVERY

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5.1 Preservation and packaging.

5.1.1 *Level A.* Cards, Aperture shall be preserved, packaged and tested in accordance with the procedures specified for the designated methods as prescribed in Specification MIL–P–116 and as described herein.

5.1.1.1 *Unit packaging.* Cards, Aperture shall be packaged Method III as follows: Each unit deck shall be placed within a close-fitting, one-piece special folding design, Type I, Class 1, fiberboard box conforming to table I of Specification PPP–B–636. When the cards do not completely fill the box, spacers, of the size of a card and fabricated of corrugated fiberboard equal to the strength of the box material, shall be placed behind the cards in sufficient numbers as to entirely fill voids, and keep cards under pressure equivalent to normal follower block pressure of a card file drawer, to prevent card curl.

5.1.2 *Level C.* Cards, Aperture shall be preserved and packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity.

5.2 Packing. Packing shall be Level A, B, or C as specified in the contract or order (see 6.1).

5.2.1 *Level A.* Five unit packages, packaged as specified in 5.1.1.1 shall be placed within a close-fitting regular slotted style, Type I, Class 1, Fiberboard box conforming ing to Specification PPP–B–636. Box closure shall be in accordance with the referenced box specification. Each fiberboard box shall be provided with Type I or Type II, Grade B, Class 2 case liner conforming to the requirements of Specification MIL–L–10547.

5.2.2 *Level B.* Five unit packages, packaged as specified in 5.1.1.1 shall be placed within a close-fitting regular slotted style, Type I, Class 1, Fiberboard box conforming to table I of Specification PPP–B–636. Closure shall be in accordance with referenced box specification.

5.2.3 *Level C.* The pack shall be as specified in 5.2.2.

5.3 Marking. Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number, and date of this specification.
(b) Types, classes, and formats of cards required (see 3.2.2 and 3.3).
(c) Delivery date and destination(s) for cards.
(d) Level of protection (see sec. 5).
(e) Corner cut (see 3.2.2).
(f) Color and position of stripe (see 3.2.2).
(g) Number of bid samples to be submitted with bid.

6.2 Definitions.

6.2.1 *Aperture card.* Aperture cards are defined in Standard MIL–STD–804.

6.3 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

(a) Type A. The total of that inspection set forth in the Quality Assurance Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Standard MIL–STD–105.

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(b) Type B. That inspection set forth in the Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced inspection provisions of Standard MIL–STD–105.

(c) Type C. A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the Quality Assurance provisions of this specification. The amount of inspection is less than that provided for in Type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

Custodians:
Army—EL
Navy—Bu Weps
Air Force—AFLC
Review:
Army—EL, MO, MU, MI
Navy—Weps, Docks
Air Force—AFLC
User:
Navy—MC

Preparing activity:
Army—EL
Project No. EDMS–0021

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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